PLamb – Void Analysis Project Thread

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Executive Summary

This document initiates the 'Void Analysis' project thread under the PLamb cosmology framework.

- The study investigates whether cosmic void environments leave detectable signatures on Supernova (SN Ia) Residuals,
- BAO locking metrics,
- Hilbert–Huang Transform (HHT),
- Intrinsic Mode Functions (IMFs),

and whether these effects support the Full Relativity (FR) and Kerr–de Sitter (KdS) models with variable speed of light (c) and accretion-driven dynamics or offer an alternative explanation.

1) Scientific Motivation

In FR and variable-c cosmologies, variations in local gravitational potential or matter density can modulate local clock-rates and energy scales.

Cosmic voids—low-density regions spanning tens of Mpc—offer an ideal setting to test these dependencies.

The hypothesis is that supernovae within or near voids show distinct $\Delta\mu(z)$ residuals, IMF2 energies, and BAO-locking strengths compared to wall or cluster environments.

2) Core Hypotheses

• H1 — Residual Shift: Mean $\Delta\mu$ of SNe within voids differs from wall SNe after standard corrections.

• H2 — IMF2 Amplification: IMF2 energy and BAO locking increase inside voids (reduced damping, partial Noether symmetry breaking).

• H3 — Phase Coherence: IMF2 frequency ridges show lower phase dispersion for void SNe.

• H4 — Entropy Coupling: IMF3 amplitude tracks environmental accretion proxies, modified by AGN scarcity in voids.

• H5 — AP/Geometry: Void shape anisotropy tests reveal deviations consistent with variable-c(z) or STF anisotropy.

• H6 — ISW-Lensing Link: Stacked ISW ΔT and lensing κ around voids correlate with IMF2 energy in corresponding redshift slices.

3) Datasets & Catalogs

Primary datasets:

• Supernovae: Pantheon+SH0ES, JWST high-z SNe.

• BAO: eBOSS DR16 (DH/rd, DM/rd) and DESI EDR for validation.

• Voids: VIDE/ZOBOV-based catalogs from SDSS/BOSS/eBOSS (Sutter et al., Mao et al.).

• CMB/ISW: Planck 2018 temperature and lensing maps.

• AGN/Environment: eROSITA and SDSS AGN indicators for entropy coupling analysis.

4) Methods Overview

Void identification uses VIDE/ZOBOV watershed algorithms ($R_{eff} \geq 10\ h^{-1}$ Mpc, $\delta_c \leq -0.7$).

SNe are matched to nearest void centers using comoving distances and redshift proximity ($|\Delta z| \leq 0.02$).

Analysis tracks include:

• T1 — $\Delta\mu$ environment contrasts.
• T2 — HHT decomposition per environment (IMF energy and BAO-locking comparison).
• T3 — BAO Locking Shift.
• T4 — ISW/Lensing Stack.
• T5 — AP Test on Voids.

5) Statistical Controls

• Phase-randomized surrogate ensembles per environment bin (≥10k; goal: 50k).

• Jackknife resampling over voids and bootstrap over SNe.

• Null tests using randomized void catalogs maintaining redshift distributions.

6) Key Metrics

- $\Delta\mu$ contrast: $\langle\Delta\mu\rangle$_void − $\langle\Delta\mu\rangle$_wall.
- IMF2 energy contrast $\Delta E_2 = E_2(\text{void}) - E_2(\text{wall})$.
- Locking strength r_2 and p_perm by environment.
- Phase dispersion $\sigma\varphi$(IMF2).
- ISW stack amplitude $\langle\Delta T\rangle$ within R_eff.
- AP parameter shift ε_AP.

7) Pipeline & Code Plan

Proposed package structure:
plamb_voids/

Modules:
io/,
match/,
hht/,
bao/,
isw/,
ap/,
stats/,
viz/.

Example CLIs:
- voids_match.py
- voids_hht.py
- voids_bao_lock.py
- voids_isw_stack.py
- voids_ap.py

8) Directory Structure

plamb_runs/voids/data/	— raw catalogs
plamb_runs/voids/matched/	— matched SN–void datasets
plamb_runs/voids/hht_env/	— IMF and surrogate outputs
plamb_runs/voids/bao_lock/	— BAO locking metrics
plamb_runs/voids/isw/	— ISW/lensing stacks
plamb_runs/voids/ap/	— AP test results
plamb_runs/voids/figs/	— figures and plots

9) Milestones

• M1: Catalog ingestion and SN–void matching.

• M2: Δμ environment contrasts.

• M3: IMF2 locking and surrogate validation.

• M4: ISW/lensing correlation tests.

• M5: AP test with variable-c predictions.

• M6: Bayesian integration and paper draft v1.

10) Immediate Next Actions

1. Prepare catalogs under plamb_runs/voids/data/.

2. Implement voids_match.py to build matched_sn_voids.csv.

3. Run CEEMDAN HHT by environment (nsur=10k).

4. Plot Δμ, IMF2 energy, and locking correlations.

5. Document environment-matching reproducibility parameters.

11) Glossary

Environment bins: interior ($r/R \leq 0.6$), rim (0.6–1.0), wall (1.0–1.5), field (>1.5).

IMF2 locking: correlation (r_2) of IMF2 frequency ridge with BAO scales (DH/rd, DM/rd).

AP parameter (ε_AP): anisotropy of void shapes along line-of-sight versus transverse direction.